One-way ANOVA and contrasts (emmeans)

Inference labs use the five-step template: Hypothesis → Visualise → Assumptions → Conduct → Conclude.

Learning objectives

- Fit a one-way ANOVA with `aov()` and interpret the omnibus F-test.
- Use `emmeans` to obtain estimated marginal means and pairwise contrasts with controlled family-wise error.
- Read a Tukey HSD output and translate it into a sentence.

Prerequisites

t-tests; linear regression basics.

Background

Analysis of variance is linear regression with a categorical predictor and a tradition. The omnibus F-test asks whether any group means differ; the follow-up contrasts ask which ones. The correct order is omnibus first, then targeted contrasts, preferably with a multiple testing correction that respects the family of tests you intended before seeing the data.

The `emmeans` package has made the post-hoc machinery substantially cleaner. `emmeans(fit, ~ group)` returns the estimated marginal means on the response scale; `pairs()` produces the pairwise differences with Tukey-adjusted p-values by default.

A common mistake is to run pairwise t-tests without correction and report the one with the smallest p-value. A less common but more insidious mistake is to run the omnibus F, see it is significant, and stop — reporting a single p-value without naming which groups differ and by how much.

Setup

```r
library(tidyverse)
library(broom)
library(emmeans)
set.seed(42)
theme_set(theme_minimal(base_size = 12))
```

1. Hypothesis

Using `ToothGrowth`, ask whether guinea-pig tooth length depends on dose of vitamin C.

Null: mean tooth length is equal across the three doses. Alternative: at least one dose differs.

2. Visualise

```r
ggplot(tg, aes(dose, len, fill = dose)) +
  geom_boxplot(alpha = 0.6, colour = "grey30") +
  labs(x = "Dose (mg/day)", y = "Tooth length") +
  theme(legend.position = "none")
```

3. Assumptions

Approximate normality within groups and equal variances.

```r
par(mfrow = c(2, 2))
plot(fit)
par(mfrow = c(1, 1))
bartlett.test(len ~ dose, data = tg)
```

4. Conduct

```r
emm <- emmeans(fit, ~ dose)
emm
pairs(emm, adjust = "tukey")
```

5. Concluding statement

> Tooth length increased with dose of vitamin C (one-way ANOVA $F =$ `round(summary(fit)[[1]]$"F value"[1], 2)`; $p =$ `signif(summary(fit)[[1]]$"Pr(>F)"[1], 2)`). Tukey-adjusted contrasts indicated that the 2 mg/day dose gave longer teeth than both lower doses (all adjusted $p < 0.05$).

Common pitfalls

- Stopping at a significant F without reporting which contrasts drive it.
- Reporting unadjusted pairwise p-values after the omnibus test.
- Treating a non-significant Bartlett test as proof of equal variances.

Further reading

- Lenth RV. *emmeans* package vignette.
- Oehlert GW. *A First Course in Design and Analysis of Experiments.*
- Maxwell SE, Delaney HD. *Designing Experiments and Analyzing Data.*

Session info

See also — chapter index

- Methods in this chapter
- Find your method (Chapter 0)